Crop Pollination Ecosystem Service Accounts

European Union, 2000 to 2024: methods and results

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Summary

We provide estimates of the crop pollination ecosystem service for the 27 Member States of the European Union in 2000, 2006, 2012, 2018 and 2024, following the accounting method of Vallecillo et al. (2018). In 2024, crops that depend on insect pollination cover 17.6 million hectares, and 53 percent of that area lies where the environment can support wild pollinators. The production attributable to the service is 14.4 million tonnes, against 12.1 million tonnes in 2000, and another 9.9 million tonnes depend on pollination but grow outside that area. At constant 2010 prices, the actual flow is worth 6.7 billion euro and the unmet demand 4.8 billion euro. Spatial crop data end in 2018, so the crop pattern of 2018 stands in for 2024.

1 Introduction

Crop pollination is the fertilisation of crops by wild insects, which maintains or increases crop production. Ecosystem accounting treats it as a service that ecosystems supply and farmers use. It therefore needs a measure of what the environment can supply, a measure of where crops need it, and an estimate of what results when the two overlap (Vallecillo et al., 2018).

This report describes how we estimated those three quantities and valued the result, why we took each decision along the way and what the estimates show. The calculations are carried out by a series of scripts in the `scripts/` folder of this repository, listed in Table 1 in the order in which they run. Each step of the Methods section carries a footnote to its script. The file `scripts/run_all.R` runs the whole series or any of its stages, and any single script can run again when only one input changes. The tables and the 100 m maps can be retrieved from the Downloads page of this site.

Table 1: Scripts of the pipeline, in the order in which they run.

Script	Purpose
01_prepare_nuts.R	Regional boundaries and a 1 km grid of regions
02_download_clc.R	Land cover at 100 m
03_prepare_catalogue.R	Insect activity index and bumblebee model at 1 km
04_prepare_capri.R	Crop area and yield at 1 km
05_prepare_roads.R	Major roads at 100 m
06_prepare_distance.R	Distance to semi-natural areas at 100 m
07_run_expert_model.R	Expert-based model at 100 m and its reduction to 1 km
08_download_eurostat.R	Crop production, production value and prices
09_write_manifest.R	List of heavy inputs, and 100 m clips of Luxembourg
10_prepare_zone_tables.R	Rasters added up by region
11_build_physical_accounts.R	Potential by region, demand, actual flow and unmet demand
12_build_monetary_accounts.R	Value of the flow and of unmet demand
13_write_maps_1km.R	Maps at 1 km and supply by ecosystem type
14_write_maps_100m.R	Maps at 100 m
15_check_maps_100m.R	Comparison of the 100 m maps with the 1 km maps

! How to read the results for 2024

Spatial crop data end in 2018, so a decision was made to use the crop area and yield of 2018 to represent 2024. Data sources with different methods and crop categories are not mixed within the series. Changes between 2018 and 2024 in demand, flow and unmet demand therefore reflect pollination potential and national statistics, and they do not reflect any relocation of crops.

i Sources of orientation

Two earlier implementations of this service provided orientation. The k.LAB project for this service (Integrated Modelling Partnership, 2026) informed the list of inputs, the table of land cover scores, the correspondence between major road categories and OpenStreetMap classes, the grouping of Eurostat crops and the public catalogue of

harmonised layers that some scripts read. The INCA plugin (European Commission, Joint Research Centre and VITO, 2026) informed the sequence of preprocessing steps and the form of the activity index. The calculations in this report are carried out by the code of this repository.

2 Methods

2.1 Overview

The idea behind the accounts is that wild pollinators forage within a limited distance of the habitat that feeds and shelters them, so a crop benefits from the service only where the surrounding environment can sustain them. The method expresses this in four components (Vallecillo et al., 2018). **Pollination potential** is a dimensionless indicator of how suitable a place is for wild pollinators. It is the mean of an expert-based model of solitary bees and a species distribution model of bumblebees, and cells with a potential of at least 0.2 form the **service providing area**. **Demand** is the area of crops that benefit from insect pollination. **Actual flow** is the production attributable to pollination where demand and the service providing area overlap, and **unmet demand** is the same quantity where they do not. Valuation then separates the share of the value of crop production that is due to the service.

The analysis covers the European territory of the EU27, on the grid of the Corine Land Cover maps in the projection EPSG:3035. Potential is modelled at 100 m and combined at 1 km, the resolution of the crop data, and the results are reported by region and mapped at both resolutions.

2.2 Territory and reporting regions

We used the NUTS 2024 boundaries (Eurostat, 2024), the European classification of territorial units, to report on 27 countries (NUTS0), 239 NUTS2 regions and 1160 NUTS3 regions. Each 1 km cell was assigned to the NUTS3 region at its centre, and coastal and border cells whose centre falls outside every region took the nearest region within 2 km, so that crop area on the coast is not lost.[1] All tables are built by adding up the NUTS3 regions, so the three levels match exactly. The Canary Islands, the Azores, Madeira and the overseas regions fall outside the climate data and are left out. The resulting grid holds 412.4 million hectares of land, of which 152.1 million are agricultural in the land cover of 2018.

[1] See `scripts/01_prepare_nuts.R`.

2.3 The expert-based model

The expert-based model scores each 100 m cell from 0 to 1 for its capacity to support wild solitary bees, in the five steps below.[2]

2.3.1 Land cover

The Corine accounting layers are the versions of the Corine Land Cover maps that the European Environment Agency prepares for comparison over time, and they are the land cover data that the method prescribes. The public catalogue that accompanies the k.LAB project (Integrated Modelling Partnership, 2026) holds version 20 for the years up to 2018, whereas the Agency published version 21 for all five years in September 2026 and marked version 20 as superseded (European Environment Agency, 2026). We therefore considered two routes.[3]

Using the harmonised catalogue layer is the shorter route, and it offers a ready-made distance map. Rebuilding from primary sources means downloading version 21 and recomputing the distance maps, and it gives one consistent product across the series, including 2024. We ran the model on both routes for 2000 and 2018 and compared them at 1 km with the same climate, road and bumblebee layers (Table 2). The two agree closely on the mean and on the area served, although a few cells change substantially. Since only the rebuilt route reaches 2024, a decision was made to use it as the baseline for all years. The comparison was run once, so the pipeline does not repeat it, and -routes keeps its result.[4]

Table 2: Pollination potential at 1 km from the rebuilt and the catalogue routes.

Year	Correlation	Mean, rebuilt	Mean, catalogue	Largest gap	Service area, rebuilt (Mha)	Service area, catalogue (Mha)	Class differs (%)
2000	0.985	0.209	0.209	0.298	228.994	229.043	2.235
2018	0.978	0.213	0.214	0.310	237.565	240.479	3.619

Each Corine class carries a score for floral availability F and for nesting suitability N, on a scale from 0 to 1, which Zulian, Maes, & Paracchini (2013) derived from the literature and an expert workshop. A cell takes the mean of the two, as Vallecillo et al. (2018) prescribe in place of the product that Zulian et al. (2013) used.

[2] See `scripts/07_run_expert_model.R`, and `scripts/helpers_pollination.R` for the equations.
[3] See `scripts/02_download_clc.R` and `scripts/03_prepare_catalogue.R`.
[4] The route is available in `scripts/07_run_expert_model.R` with `routes <- "catalogue"`, which needs the catalogue layers that `scripts/03_prepare_catalogue.R` retrieves.

$$S = \frac{F + N}{2} \tag{1}$$

The scores are those of Table S1 in the supplement to Zulian et al. (2013), shown in Table 16 in the appendix. That article refines them with crop maps, a high resolution forest map, high nature value farmland, riparian areas and road sides, whereas the method of Vallecillo et al. (2018) works from Corine alone, so a decision was made to use the values that the supplement offers for readers who rely on Corine data only. These include an average for olive groves, which the supplement scores as 0.2 and 0.4 when intensive and 0.6 and 0.6 when extensive. We took the table from the k.LAB project (Integrated Modelling Partnership, 2026) and compared it with the supplement, and all 43 classes agree.

2.3.2 Major roads

Major roads cannot provide food or nesting sites, so cells on a major road score zero. The method names a commercial road network, TeleAtlas, which is proprietary. A decision was made to use OpenStreetMap (OpenStreetMap contributors, 2026) instead, taking motorways, trunk roads and primary roads with their links, which correspond to the same categories, and rasterising them on the 100 m grid.[5] As in the method, the road map is a single snapshot that applies to all years.

2.3.3 Forest edges

Wild bees use forest margins more than forest interiors. Following Vallecillo et al. (2018), the outer ring of 100 m of each forest patch keeps its score and the interior scores zero. Forest means the broad-leaved, coniferous and mixed forest classes 311, 312 and 313, and a forest cell belongs to the interior when all eight of its neighbours are also forest. The edge keeps the Corine score of its class from Table S1. The supplement to Zulian et al. (2013) also holds separate scores for forest edges in its Table S3, which are 0.1 higher for both floral availability and nesting suitability, and which it applies to an edge of 50 m.

2.3.4 Distance to semi-natural areas

Pollinator richness and visits decline exponentially with isolation from natural habitat, at a mean rate of 0.00046 per metre across 23 studies (Garibaldi et al., 2011; Ricketts et al., 2008). The method applies this decay to the score, using the distance d in metres from each cell to the nearest semi-natural area.

[5] See `scripts/05_prepare_roads.R`.

$$S_d = \exp\left(\ln(S) - 0.00046\, d\right) = S\, e^{-0.00046\, d} \tag{2}$$

The score is zero beyond 20 km. Semi-natural areas are the classes 311, 312, 313, 321, 322, 323, 324, 331 and 333. We computed the Euclidean distance between cell centres for each of the five land cover maps.[6]

2.3.5 Activity of insects

Habitat is of no use if it is too cold for insects to leave the nest. The activity index A of Corbet et al. (1993), as adapted by Zulian et al. (2013), expresses in percent how active insects are as a function of air temperature T in degrees Celsius and solar irradiance R in watts per square metre, through the temperature of a black globe that simulates the body of an insect.

$$A = -39.3 + 4.01\, T_b, \qquad T_b = -0.62 + 1.027\, T + 0.006\, R \tag{3}$$

The method names the Agri4Cast climate grid, whose access requires a sign-in that was not available. A decision was made to use instead the ERA5 temperature (Hersbach et al., 2023) and the Copernicus radiation (Copernicus Climate Change Service, 2020) of the public catalogue, which reach 2025 and whose radiation is already expressed over day length. The method text refers to minimum temperature and its table of inputs to mean temperature, and a decision was made to use mean temperature, which is the variable in Zulian et al. (2013). For each reporting year we took the three years centred on it, with 2023 to 2025 for 2024, and computed the index for each month from April to September. Each month was limited to the range from 0 to 100, as Zulian et al. (2013) do, before averaging the six months and interpolating the result bilinearly to 1 km and then to 100 m.[7]

The expert-based model E is the product of the decayed score and the index.

$$E = S_d\, \frac{A}{100} \tag{4}$$

We ran the five steps over the whole grid in tiles, for each year, and reduced the result to 1 km with the mean of the land cells of each kilometre. The reduction is what the following steps read, and the 100 m tiles remain outside the repository, listed with their source, size and checksum in a manifest.[8] To let a reader inspect the 100 m chain with no download, the repository also holds the 100 m inputs of Luxembourg in 2018. Running the model on them reproduces the full run with a largest difference of 5e-05 in any kilometre inside the country.[9]

[6] See `scripts/06_prepare_distance.R`.
[7] See `scripts/03_prepare_catalogue.R`.
[8] See `scripts/09_write_manifest.R`.
[9] See the function `run_pilot_luxembourg()` in `scripts/helpers_accounts.R`.

2.4 The bumblebee model

The second model estimates where bumblebees occur. The method fits a species distribution model to records of 47 species and projects it to each period, which requires the fitted models. These are unavailable, so a decision was made to use the published aggregated layer of Polce et al. (2018), which gives the mean probability of occurrence of the 47 species at 10 km. We interpolated it bilinearly to 1 km, and land cells left without a value along the coast took the mean of their neighbours within 10 km.[10] Its values run from 0.04 to 0.73 on land. The layer does not change over time, so all change in potential over the series comes from land cover and climate.

2.5 Pollination potential and the service providing area

The potential P of a 1 km cell is the mean of the expert-based model and the bumblebee model.

$$P = \frac{E + B}{2} \tag{5}$$

The classes follow Vallecillo et al. (2018), namely high above 0.3, medium from 0.2 to 0.3, low from 0.1 to 0.2 and none below 0.1. Cells of medium and high potential form the service providing area, where pollinators are assumed to be present. We added up land and agricultural land in each class by NUTS3 region.[11]

2.6 Demand

Demand is the area of the ten crop groups in Table 3, which the method takes from the CAPRI agricultural model disaggregated to 1 km. The method had two years of these data. We used the later series of Koeble et al. (2024), which covers 2000 to 2018 and offers the exact reporting years. The layers were read from the public catalogue, which holds the same variables for those years, and reduced to one compact stack of area and one of yield per year.[12]

[10] See `scripts/03_prepare_catalogue.R`.

[11] See `scripts/10_prepare_zone_tables.R`.

[12] See `scripts/04_prepare_capri.R`. The script `scripts/04b_check_capri.R` compares these layers with the original archives. That comparison has not been run, because the archives are very large.

Table 3: Crop groups, dependency on insect pollination (Klein et al., 2007) and items of the economic accounts for agriculture (Vallecillo et al., 2018).

Code	Crop	Dependency ratio	Class	Eurostat items
APPL	Apples, pears and peaches	0.650	High	AM061100, AM061200, AM061300
OFRU	Other fruits	0.400	Moderate	AM061900
RAPE	Rapeseed	0.250	Moderate	AM021100
SUNF	Sunflower	0.250	Moderate	AM021200
SOYA	Soya	0.250	Moderate	AM021300
OOIL	Other oilseeds	0.175	Moderate	AM021900
PULS	Pulses	0.050	Low	AM022000
TEXT	Flax and hemp	0.050	Low	AM029100
TOMA	Tomatoes	0.050	Low	AM041200
CITR	Citrus fruits	0.050	Low	AM062000

Adding up the layers showed that some kilometres hold more than the 100 hectares a square kilometre can contain, with 2,395 such cells in 2018 and a largest value of 1,023 hectares. These values reflect multiple cropping and the way the source allocates reported land. They hold 1.8 percent of demand in that year, and truncating them would understate the area, so a decision was made to keep them.

2.6.1 Countries outside the crop data

The method notes that CAPRI does not cover Croatia, Malta and Cyprus. Reading the later series gives a more nuanced picture (Table 4). Croatia is absent, with a few hectares that come from cells along its border. Cyprus and Malta, by contrast, do hold crop area, and in Cyprus it is close to the national statistics. A decision was made to treat the three countries in the same way, in line with the method, so they stay out of the crop maps and take the estimate described in Section 2.7.3. Cyprus and Malta together hold 0.05 percent of the actual flow in 2024, so the choice has little effect on the totals.

Table 4: Crop area in the three countries in 2018, in hectares, and the actual flow estimated for them.

Country	CAPRI series (ha)	Eurostat (ha)	Estimated flow (t)
Croatia	30	202,780	113,389
Cyprus	8,172	8,770	6,438
Malta	466	0	1,546

2.7 Actual flow and unmet demand

The actual flow of crop c in cell i is its production times its dependency ratio δ_c (Klein et al., 2007) when the cell belongs to the service providing area Ω. Unmet demand applies the same product to the cells outside it, so flow and unmet demand add up to all production that depends on pollination.

$$F_{c,i} = a_{c,i}\, y_{c,i}\, k_c\, \delta_c \quad \text{if } i \in \Omega, \qquad U_{c,i} = a_{c,i}\, y_{c,i}\, k_c\, \delta_c \quad \text{if } i \notin \Omega \tag{6}$$

Here a is the crop area, y the yield and k a scaling factor for the crop and country, explained below. A decision was made to measure unmet demand in tonnes by analogy with the flow, because the method defines it by area only and the monetary accounts need a physical quantity. We report the area of met and unmet demand as well.

2.7.1 Scaling to national statistics

The method anchors the accounts in official statistics, so we compared the CAPRI production of each crop and country with the national production of Eurostat, in the three-year mean around each reporting year (Eurostat, 2026a).[13] Table 5 shows that the two sources agree well for most crops, with two exceptions that matter. CAPRI places about twice the Eurostat production of other fruits, and about 3 times the Eurostat area of flax and hemp.

[13] See `scripts/08_download_eurostat.R`.

Table 5: Crop area and production in CAPRI and Eurostat in 2018, before scaling, for the crops and countries that both report.

Crop	CAPRI (kha)	Eurostat (kha)	Area ratio	CAPRI (kt)	Eurostat (kt)	Production ratio
Apples, pears and peaches	522.2	823.4	0.63	13,260.6	18,038.0	0.74
Citrus fruits	494.9	494.9	1.00	10,762.4	10,719.1	1.00
Other fruits	2,183.8	1,951.8	1.12	15,076.5	8,601.6	1.75
Other oilseeds	470.6	249.7	1.88	796.4	1,240.1	0.64
Pulses	2,113.5	2,181.7	0.97	3,703.7	4,449.9	0.83
Rapeseed	6,282.5	5,824.0	1.08	18,033.7	17,498.5	1.03
Soya	872.1	861.7	1.01	2,486.5	2,590.7	0.96
Sunflower	3,961.4	4,186.7	0.95	9,990.0	10,126.1	0.99
Flax and hemp	495.1	157.7	3.14	1,197.8	958.5	1.25
Tomatoes	237.9	242.9	0.98	16,958.8	17,139.8	0.99

A decision was therefore made to keep the CAPRI pattern of each crop within a country and scale its production to the Eurostat total, with the factor k_c = Eurostat production/CAPRI production. Maps and tables then agree with official totals. The national pollination contribution, which is the actual flow divided by total production, is unaffected by this factor, so it remains the production-weighted share that Vallecillo et al. (2018) define. Demand stays in the hectares of CAPRI, since the crop pattern on the map comes from that source. Where Eurostat has no figure for a crop and country, the CAPRI production is kept and flagged. There are 10 such cases, the largest being tomatoes in Portugal in 2000 and 2006, with about 1.2 million tonnes in 2006.[14]

2.7.2 Cases where the sources disagree on the crop

Applying the factors exposed a set of crops and countries for which the two sources disagree on whether the crop is present at all. CAPRI places 233,507 tonnes of other fruits in Luxembourg in 2000, against 1,000 tonnes in Eurostat, and about 261 thousand tonnes of flax and hemp in Greece in 2006, where Eurostat reports none. Conversely, Eurostat reports 5,045 tonnes of rapeseed in Greece in 2000, where CAPRI has almost nothing. Scaling such cases would place

[14] See `scripts/11_build_physical_accounts.R`, where these calculations are carried out.

unrealistic amounts in a few cells, so a decision was made to set them aside when the factor is above 10 or below 0.1. Table 6 shows that they are numerous but small, at under one percent of production in every year. Production that Eurostat reports for these cases moves to the estimate below, and CAPRI production that Eurostat does not report is dropped.

Table 6: Crops and countries set aside because CAPRI and Eurostat differ by more than a factor of ten.

Year	Cases	CAPRI production set aside (kt)	Eurostat production re-estimated (kt)	Share of all production (%)
2000	30	826.7	280.1	0.4
2006	25	524.5	5.4	0.0
2012	27	555.4	7.8	0.0
2018	27	338.6	63.7	0.1
2024	30	342.4	299.8	0.3

2.7.3 Production without a place on the map

Some national production has no place on the map, either because of the countries discussed above or because of the cases just set aside. For these, the share of the agricultural land of each region inside the service providing area stands in for the share of the crop inside it, and the agricultural land of each region shares out the national figure. Area follows the same shares, from the national area in Eurostat. Malta reports production but no area until 2024, so it shows flow with no hectares of demand in earlier years. These values are flagged as estimates in the tables and do not enter the crop maps. They hold 0.7 to 1.3 percent of the flow, depending on the year.

2.8 Monetary valuation

Pollination contributes to a product that the national accounts already record, namely crops. The method therefore does not price the service separately. It takes the value of production in the economic accounts for agriculture and separates the part due to pollination, so no value is counted twice (Vallecillo et al., 2018). We retrieved the production value at basic price in the prices of the previous year from Eurostat dataset `aact_eaa02` (Eurostat, 2026b), as the mean of the three years around each reporting year, and we linked it to the crop groups through the correspondence in Table 3.[15]

[15] See `scripts/08_download_eurostat.R`.

The pollination contribution C of a crop in a country is its actual flow over its total production T, and the value of the flow is the production value V times that contribution. The value of unmet demand follows in the same way.

$$C = \frac{F}{T}, \qquad V_F = V\,C, \qquad V_U = V\,\frac{U}{T} \tag{7}$$

Values in the prices of the previous year still drift with inflation across a series of 24 years, so a decision was made to express them in constant prices of 2010, as the method does not prescribe a base for a longer series. We multiplied them by the deflator $H_{2010}/\bar{H}_y$, where H is the harmonised index of consumer prices of the European Union (Eurostat, 2026c) and $\bar{H}_y$ is its mean over the three years that the prices of each reporting year refer to. The deflator runs from 1.27 for 2000 to 0.74 for 2024.[16]

The two sources do not match perfectly. A production value in a country with no physical production has no contribution and receives no value, which leaves out 2,555 million euro across all years and countries. A country that reports production of a crop but no value for it has flow with no monetary counterpart, which concerns 0.2 percent of the physical flow. Regions receive the national value of each crop in proportion to their physical flow and unmet demand, through a value per tonne that is uniform within a crop and country.

2.9 Maps

We wrote the maps of potential, demand, actual flow, unmet demand and value at 1 km from the rasters and the factors and prices of the tables, and then read their totals back from disk to compare them with the tables.[17] The 1 km maps are a working product, kept outside the repository, from which the published 100 m maps are built.

The 100 m maps add spatial detail in two ways.[18] Potential is computed directly as the mean of the 100 m expert-based model and the bumblebee model interpolated to 100 m. The other quantities exist only at 1 km, so they are allocated within each kilometre, and each kilometre keeps its total. Demand, unmet demand and the flow and value on agroecosystems go to agricultural cells, and the flow for all ecosystem types goes to all land cells in proportion to their expert-based suitability. A kilometre with no positive weight spreads its value evenly over its land cells. The detail within each kilometre therefore comes from land cover, and crops are not observed at that scale. To make the files several times smaller, values are stored as whole numbers in small units, namely potential multiplied by 10 000, demand in square metres, flow and unmet demand in kilograms, and value in euro.

[16] See `scripts/12_build_monetary_accounts.R`, where these calculations are carried out.
[17] See `scripts/13_write_maps_1km.R`.
[18] See `scripts/14_write_maps_100m.R`.

To describe which ecosystems support the flow, we shared the flow of each kilometre among ecosystem types in proportion to their share of its expert-based suitability, and we attributed to cropland the small amount of flow in cells with no support.[19]

2.10 Checks

The scripts stop if any check on the tables or on the 1 km maps fails. Table 7 records the largest relative gap of each check on the tables. The value of the flow in any crop and country also never exceeds the production value times the dependency ratio.

Table 7: Consistency checks of the tables, with the largest relative gap and the tolerance that would stop the scripts.

Check	Largest gap	Tolerance
Flow by NUTS3 against NUTS0	0.0e+00	1e-04
Flow by NUTS2 against NUTS0	0.0e+00	1e-04
Flow plus unmet demand against dependent production	0.0e+00	1e-04
National production against Eurostat	2.2e-16	1e-03
Value of flow, regions against countries	0.0e+00	1e-04
Value of unmet demand, regions against countries	1.6e-16	1e-04
Value of flow by NUTS3 against NUTS0	0.0e+00	1e-04

The maps agree with the tables once the estimates, which the maps leave out, are set aside. In 2024 the flow maps add up to 14.18 million tonnes, against 14.18 million tonnes in the tables without estimates and 14.36 million tonnes with them. We also read the 100 m maps back from disk and compared them with the 1 km maps for all five years.[20] No kilometre differs by more than the 50 units that rounding allows, and the totals of demand, flow, unmet demand and value stay within 0.03 percent of the 1 km totals.

Potential differs by at most 0.053 on the scale from 0 to 1 in any kilometre, which comes from interpolating the bumblebee model to 100 m rather than to 1 km. The mean of the map of potential on agricultural land is about 5 percent below that of its 1 km map because the two averages weigh cells differently. The 100 m mean counts each agricultural cell, and the 1 km mean counts each kilometre once however little agricultural land it holds.

[19] See the function `supply_by_ecosystem()` in `scripts/helpers_accounts.R`.
[20] See `scripts/15_check_maps_100m.R`.

3 Results

3.1 Pollination potential

The environment became somewhat more suitable for wild pollinators over the series (Table 8 and Figure 1). Land of high potential grew from 24.4 to 37.2 million hectares, mostly at the expense of land of low potential, while land of medium potential stayed close to 200 million hectares. Most of the rise seems to come from climate. The activity index rose from a mean of 31.1 percent to 35.3 percent, by about the same proportion as the expert-based model, which multiplies it. This reading rests on the parallel between the two series, as we did not separate the effects of climate and land cover.

Table 8: Land by class of potential, in million hectares, and the mean of each layer on land.

Measure	2000	2006	2012	2018	2024
High potential (Mha)	24.4	26.9	27.8	32.6	37.2
Medium potential (Mha)	201.6	203.2	201.7	201.8	202.6
Low potential (Mha)	173.1	169.1	169.7	165.0	160.1
No potential (Mha)	13.3	13.1	13.2	12.9	12.4
Mean potential	0.209	0.211	0.211	0.213	0.216
Mean of the expert-based model	0.092	0.096	0.096	0.101	0.106
Mean of the bumblebee model	0.325	0.325	0.325	0.325	0.325
Mean activity index (%)	31.1	32.3	32.4	34.2	35.3

3.2 Demand, actual flow and unmet demand

Demand grew from 13.8 to 17.6 million hectares, and actual flow from 12.1 to 14.4 million tonnes (Table 9). The share of demand inside the service providing area changed much less,

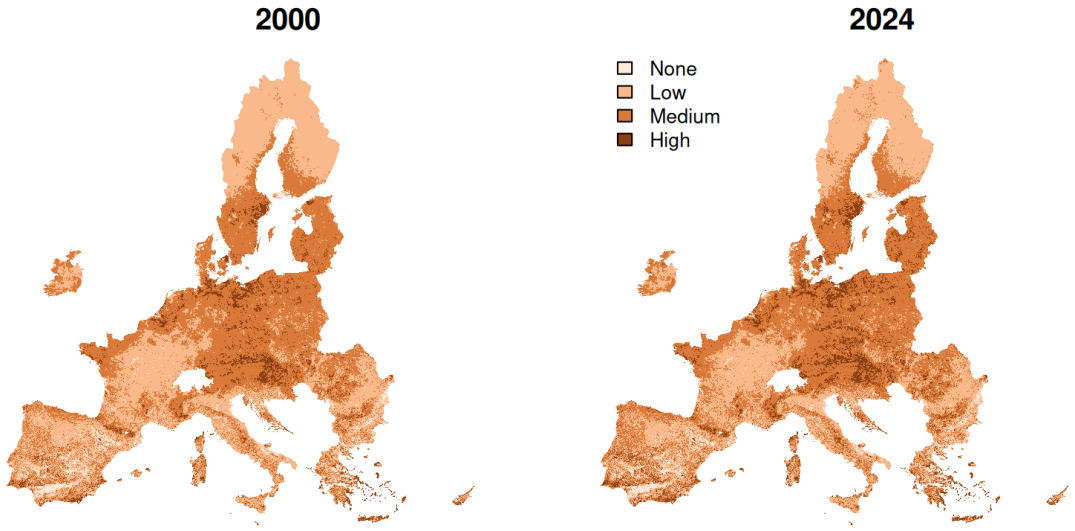


Figure 1: Pollination potential by class, 2000 and 2024.

from 48.2 to 53.0 percent. Growth in flow therefore comes mainly from more dependent crops and higher production, and only in part from better conditions for pollinators.

Table 9: Demand, actual flow and unmet demand in the EU27. Demand and flow of 2024 use the crop area and yield of 2018.

Year	Demand (Mha)	Met demand (%)	Actual flow (Mt)	Unmet demand (Mt)	Share of flow estimated (%)
2000	13.8	48.2	12.1	9.0	1.3
2006	14.7	51.7	12.2	9.2	0.7
2012	14.7	52.7	12.8	9.2	0.8
2018	17.5	52.3	14.1	10.7	0.9
2024	17.6	53.0	14.4	9.9	1.2

Two features of the series deserve attention. Between 2012 and 2018, demand rises by 2.8 million hectares, and about 47 percent of that rise comes from other fruits, which fall to 0.9 million hectares in 2012 against about 2 million in the other years. Both CAPRI and Eurostat show this dip, which is more likely to reflect gaps in the reported data than a real change. Between 2018 and 2024, with the crop pattern held fixed, unmet demand falls from 10.7 to

9.9 million tonnes. Part of this is the growth of the service providing area, and part is lower national production, notably of sunflower, whose unmet demand falls from 1.64 to 1.32 million tonnes. It should not be read as crops moving towards better habitat.

Figure 2 shows where this happens in 2024. Demand spreads across most of the farmland of the continent. Actual flow concentrates in Poland, Germany, Hungary, the Benelux and the north of France, and unmet demand in central and southern France, Italy, Spain and Romania.

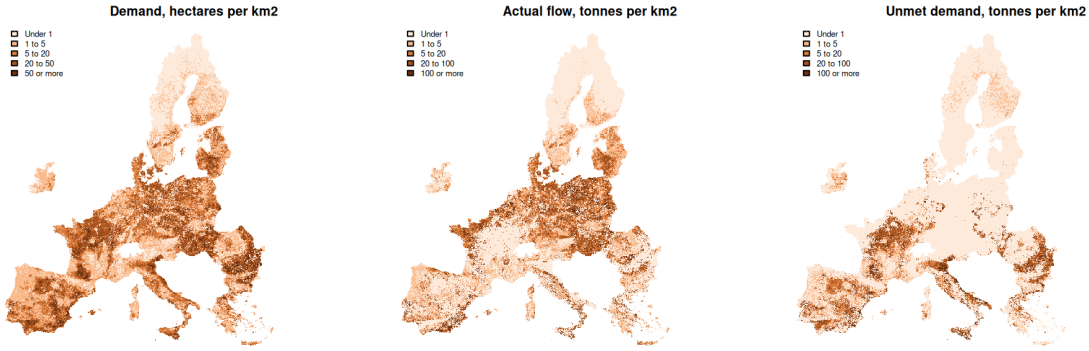


Figure 2: Demand, actual flow and unmet demand at 1 km in 2024, which uses the crop area and yield of 2018.

In the service providing area, most demand lies at medium potential. In 2006, 6 percent of the area of CAPRI crops on the map lies at high potential and 45 percent at medium potential, which is close to the 6 and 44 percent that Vallecillo et al. (2018) report for that year, and the remaining 49 percent is close to their 50 percent of unmet demand.

3.3 Crops

Fruit dominates the service (Table 10). Apples, pears, peaches and other fruits make up 69 percent of the value of the flow in 2024, and rapeseed another 18 percent. Tomatoes and citrus illustrate the role of the dependency ratio, since they hold large production and little flow because only 5 percent of their production depends on pollination. Sunflower and soya have the lowest shares of met demand among the crops of moderate dependency.

Table 10: Accounts by crop in the EU27 in 2024, which use the crop area and yield of 2018. Value is in million euro at constant 2010 prices.

Crop	Demand (kha)	Met (%)	Production (kt)	Flow (kt)	Unmet (kt)	Contribution	Value of flow	Value of unmet
Apples, pears and peaches	528	75	17,214	7,222	3,967	0.42	2,442	1,750
Other fruits	2,217	55	8,846	1,962	1,577	0.22	2,218	1,506
Rapeseed	6,300	63	18,748	3,069	1,618	0.16	1,242	557
Sunflower	4,024	36	8,900	901	1,324	0.10	347	520
Tomatoes	239	36	16,533	335	492	0.02	187	148
Soya	1,006	42	2,909	313	414	0.11	117	145
Citrus fruits	500	43	10,588	228	301	0.02	82	106
Pulses	2,120	55	5,005	167	83	0.03	52	32
Other oilseeds	477	69	1,075	136	52	0.13	39	10
Flax and hemp	157	58	1,139	32	25	0.03	15	21

3.4 Countries

The share of met demand separates the continent sharply (Table 11). It is above 80 percent across most of central and northern Europe, and it is far lower in several of the countries with the largest demand, with 25 percent in Italy, 28 percent in France, 34 percent in Spain and 24 percent in Romania. As a result, Poland has the largest value of flow, at 1,020 million euro, while France has the largest value of unmet demand, at 1,180 million euro (Figure 3). In Italy, the value of unmet demand is about twice the value of the flow. The Netherlands stands out for the opposite reason, with 374 million euro from only 29 thousand hectares, which reflects a high value per hectare and a high share of met demand.

Table 11: Accounts by country in 2024, which use the crop area and yield of 2018. Value is in million euro at constant 2010 prices. An asterisk marks countries with estimated values.

Country	Demand (kha)	Met (%)	Flow (kt)	Unmet (kt)	Value of flow	Value of unmet
PL*	1,589	91	3,523	267	1,020	86
FR	2,926	28	1,424	2,113	966	1,180
ES*	2,596	34	1,460	1,795	753	878
DE*	1,525	90	1,651	213	691	90
IT*	1,199	25	1,323	2,618	473	904
EL*	517	51	565	731	404	608
NL	29	92	417	37	374	33
RO	2,105	24	600	1,059	353	465
HU	1,160	84	819	138	347	58
PT	250	52	277	277	251	236
BE	44	96	405	5	234	3
AT	192	100	295	0	163	0
CZ	556	94	361	23	140	9
LT*	445	99	269	2	111	1
DK	181	89	189	23	91	11
BG*	1,158	32	190	369	77	146
SK	312	64	143	81	60	33
SE	149	98	116	3	58	1
LV	162	100	97	0	40	0
FI*	84	84	27	6	37	10
HR*	215	64	109	62	35	20
SI*	14	92	32	5	28	4
EE	123	99	43	0	17	0
IE	20	53	19	20	11	14
CY*	9	46	6	7	3	4

Country	Demand (kha)	Met (%)	Flow (kt)	Unmet (kt)	Value of flow	Value of unmet
LU*	4	76	3	1	2	0
MT*	0	96	1	0	1	0

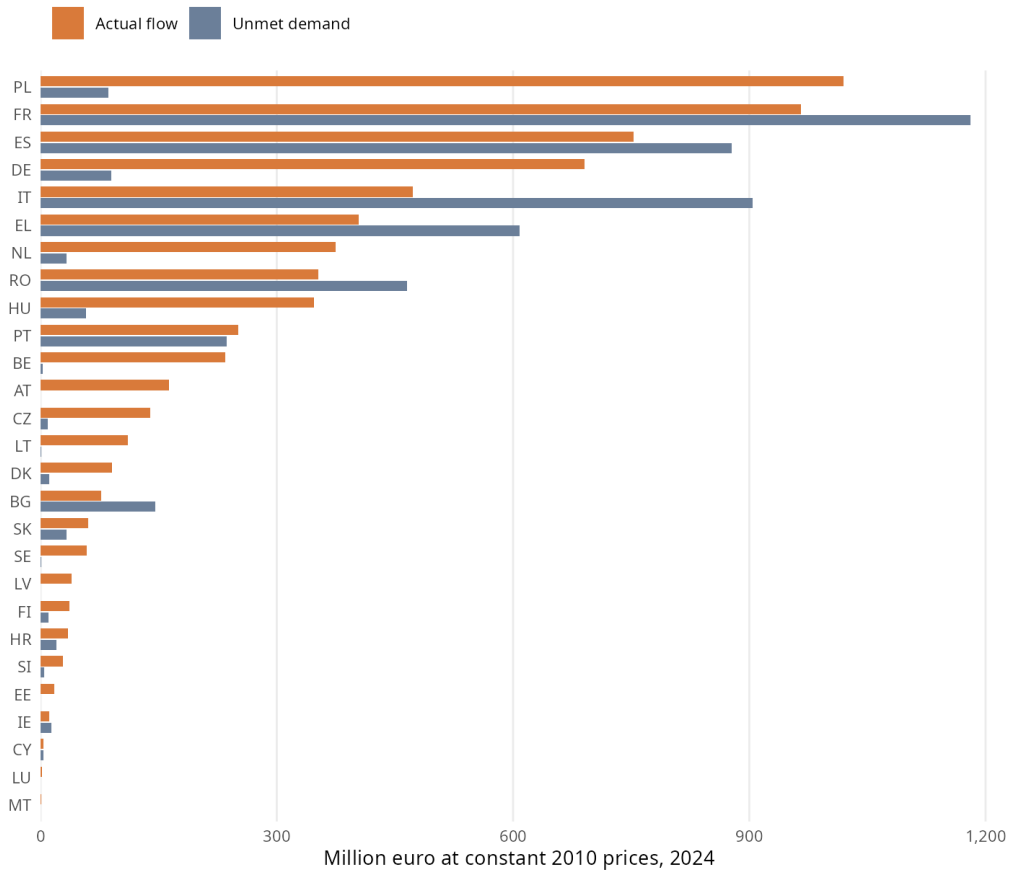


Figure 3: Value of the actual flow and of unmet demand by country, 2024, which uses the crop area and yield of 2018.

3.5 Value over time

The value of the actual flow at constant 2010 prices rose from 5.2 to 6.7 billion euro between 2000 and 2024 (Table 12). Most of the much larger rise in the prices of the previous year, from 4.1 to 9.1 billion euro, is general inflation.

Table 12: Value of dependent crops, of the actual flow and of unmet demand in the EU27. Values of 2024 use the crop area and yield of 2018.

Year	Dependent crops, value (M EUR 2010)	Value of flow (M EUR 2010)	Value of unmet demand (M EUR 2010)	Value of flow (M EUR, prices of previous year)
2000	32,785	5,236	4,324	4,116
2006	31,414	5,248	4,148	4,708
2012	34,723	5,978	4,311	6,146
2018	37,448	6,323	4,655	6,921
2024	40,594	6,741	4,794	9,054

3.6 Ecosystems that support the flow

Cropland itself supports 51 percent of the mapped flow in 2024, followed by woodland and forest with 27 percent (Table 13). This follows from the expert-based scores and from how much cropland surrounds the crops, and it does not observe where pollinators come from.

Table 13: Mapped actual flow by ecosystem type, in thousand tonnes. Flow of 2024 uses the crop area and yield of 2018.

Ecosystem	2000	2006	2012	2018	2024
Cropland	5,613	5,919	6,411	7,067	7,296
Woodland and forest	3,160	3,085	3,268	3,718	3,771
Grassland	1,428	1,409	1,416	1,515	1,492
Urban	583	625	672	727	763
Heathland and shrub	979	908	776	784	721
Sparsely vegetated land	174	105	79	74	73
Wetlands	52	58	63	71	69
Rivers and lakes	0	0	0	0	0

3.7 Comparison with the published accounts

Vallecillo et al. (2018) report for 2006 an actual flow of 13 million tonnes, 15 percent of the production of dependent crops, for the EU-28. We obtain 12.2 million tonnes, or 15 percent, for the EU27, which is a close match. National contributions are close in Germany, France and Poland, as Table 14 shows for two crops, and they differ more in Italy, Spain and Romania. The comparison serves for orientation only, because the crop data, the land cover version and the climate source differ.

Table 14: Pollination contribution in 2006 in this report and in Vallecillo et al. (2018), Table 4.5.

Country	Pome fruit, here	Pome fruit, published	Rapeseed, here	Rapeseed, published
DE	0.56	0.57	0.22	0.21
ES	0.31	0.26	0.06	0.03
FR	0.27	0.29	0.05	0.05
IT	0.22	0.13	0.06	0.05
PL	0.61	0.59	0.22	0.22
RO	0.40	0.47	0.03	0.05

The value differs by more, and the reason lies mostly in one crop (Table 15). In 2006, in the same prices of the previous year, we obtain 4.7 billion euro against the published 3.1 billion euro. The production value of most crops is close to the published one, despite the different coverage of EU27 and EU-28. Apples, pears and peaches are the main exception, at 6,620 million euro against 2,241 million euro. The correspondence of the method assigns three Eurostat items to this crop group, and a decision was made to add up the three as it prescribes. This one difference explains about 88 percent of the gap in the value of the flow. Flax and hemp also differ, at 390 against 1,135 million euro, but they weigh little in the flow.

Table 15: Production value and pollination contribution in 2006, for the EU27 in this report and the EU-28 in Vallecillo et al. (2018), Tables 4.4 and 4.5. Value is in million euro at the prices of the previous year.

Crop	Value, here	Value, published	Contribution, here	Contribution, published	Flow value, here
Apples, pears and peaches	6,620	2,241	0.38	0.48	2,487

Crop	Value, here	Value, published	Contribution, here	Contribution, published	Flow value, here
Other fruits	5,494	5,923	0.23	0.30	1,208
Rapeseed	3,670	4,177	0.17	0.17	591
Tomatoes	5,974	5,928	0.02	0.04	148
Sunflower	1,585	1,585	0.09	0.12	140
Citrus fruits	3,302	3,238	0.02	0.02	67
Soya	287	287	0.08	0.16	23
Other oilseeds	136	155	0.07	0.12	20
Pulses	724	884	0.03	0.03	18
Flax and hemp	390	1,135	0.02	0.04	7

4 Limitations

Potential. Potential is an indicator of environmental suitability and does not count pollinators or visits to flowers. The model does not include pesticides or fertilisers, for which data are lacking across time and space, and it covers wild solitary bees and bumblebees only. The land cover scores are expert judgements, and the method does not attach uncertainty to them. Using the separate edge scores of Table S3 in Zulian et al. (2013) would raise the suitability of forest edges by 0.1, an effect that we have not measured. The service providing area is a threshold, so a small change in potential can move a cell in or out of it. The bumblebee layer is static and the road map is one snapshot, so neither contributes to change over time, and the rise in potential leans on climate.

Crops. Demand and yield come from a model that disaggregates regional data to 1 km, so the location of a crop within a region is an estimate. The crop area and yield of 2018 stand in for 2024, which is the largest assumption in the series. Demand is reported in the hectares of CAPRI, which for flax and hemp are about three times those of Eurostat, although this crop group weighs little in the flow. The dip in other fruits in 2012 affects every comparison with that year. Dependency ratios are fixed literature values, and more recent reviews propose different values for some crops.

Flow. Actual flow treats a crop as pollinated when it grows in the service providing area, with no measure of how much pollination it receives. Changes in flow also reflect changes in production for reasons other than pollination, so change in the area of met demand is the better guide to change in the service. Values for Croatia, Malta and Cyprus, and for the few other crops without a place on the map, are estimates that assume the crop is spread across

the agricultural land of the country. Cyprus and Malta could enter the crop maps in future, since the later CAPRI series holds area for them.

Value. The valuation attributes to pollination a share of the value of production at basic price, without removing the costs of farmers, so it measures a contribution to value and does not measure a rent. A consumer price index converts the values to 2010 prices, so the constant series removes general inflation and leaves changes in relative crop prices in place. Value per tonne is uniform within a crop and country, so maps of value inherit the physical pattern. Other valuation methods exist (Vallecillo et al., 2018), and they would give different figures.

Maps at 100 m. These maps keep the total of each kilometre, and they are not meant to be read cell by cell. Where the crop data place crops in a kilometre in which Corine shows few agricultural cells, those cells receive the whole value, so a single 1 hectare cell can show more area or tonnes than a hectare could hold. Sums over areas of a kilometre or more are the reliable reading.

Appendix: land cover scores

Table 16: Floral availability and nesting suitability by Corine class, from Table S1 in the supplement to Zulian et al. (2013). Olive groves take the average of the intensive and extensive scores.

Corine code	Class	Floral	Nesting	Ecosystem
111	Continuous urban fabric	0.05	0.10	Urban
112	Discontinuous urban fabric	0.30	0.30	Urban
121	Industrial or commercial units	0.05	0.10	Urban
122	Road and rail networks and associated land	0.25	0.30	Urban
123	Port areas	0.00	0.30	Urban
124	Airports	0.10	0.30	Urban
131	Mineral extraction sites	0.05	0.30	Urban
132	Dump sites	0.00	0.05	Urban

Corine code	Class	Floral	Nesting	Ecosystem
133	Construction sites	0.00	0.10	Urban
141	Green urban areas	0.25	0.30	Urban
142	Sport and leisure facilities	0.05	0.30	Urban
211	Non-irrigated arable land	0.20	0.20	Cropland
212	Permanently irrigated land	0.05	0.20	Cropland
213	Rice fields	0.05	0.20	Cropland
221	Vineyards	0.60	0.40	Cropland
222	Fruit trees and berry plantations	0.90	0.40	Cropland
223	Olive groves	0.40	0.50	Cropland
231	Pastures	0.20	0.30	Grassland
241	Annual crops associated with permanent crops	0.50	0.40	Cropland
242	Complex cultivation patterns	0.40	0.40	Cropland
243	Land principally occupied by agriculture, with significant areas of natural vegetation	0.75	0.70	Cropland
244	Agro-forestry areas	0.50	1.00	Cropland
311	Broad-leaved forest	0.90	0.80	Woodland and forest
312	Coniferous forest	0.30	0.80	Woodland and forest
313	Mixed forest	0.60	0.80	Woodland and forest
321	Natural grasslands	1.00	0.80	Grassland
322	Moors and heathland	1.00	0.90	Heathland and shrub

Corine code	Class	Floral	Nesting	Ecosystem
323	Sclerophyllous vegetation	0.75	0.90	Heathland and shrub
324	Transitional woodland-shrub	0.85	1.00	Woodland and forest
331	Beaches, dunes, sands	0.10	0.30	Sparsely vegetated land
332	Bare rocks	0.00	0.00	Sparsely vegetated land
333	Sparsely vegetated areas	0.35	0.70	Sparsely vegetated land
334	Burnt areas	0.20	0.30	Sparsely vegetated land
335	Glaciers and perpetual snow	0.00	0.00	Sparsely vegetated land
411	Inland marshes	0.75	0.30	Wetlands
412	Peat bogs	0.50	0.30	Wetlands
421	Salt marshes	0.55	0.30	Wetlands
422	Salines	0.00	0.00	Wetlands
423	Intertidal flats	0.00	0.00	Wetlands
511	Water courses	0.00	0.00	Rivers and lakes
512	Water bodies	0.00	0.00	Rivers and lakes
521	Coastal lagoons	0.00	0.20	Rivers and lakes
522	Estuaries	0.00	0.00	Rivers and lakes

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